



TO WHOMSOEVER IT MAY CONCERN

I am writing to enthusiastically recommend Mr. Abhisek Sen for any future opportunities that may come his way. He worked under my supervision as an intern in my Autonomous Systems Laboratory at IIITDM Kancheepuram during May-July 2024, and I had the pleasure of witnessing his exceptional talent and dedication firsthand.

During his internship, Abhisek was involved in a project with a problem statement - **Autonomous Navigation of a Ground Robot with LiDAR and Vision based Simultaneous Localization and Mapping (SLAM)**. In this project he accomplished the following tasks:

- **ROS2 Integration with Velodyne VLP-16 LIDAR:** Successfully integrated ROS2 with Velodyne VLP-16 LIDAR to enable real-time environment sensing and navigation capabilities.
- **Point Cloud Data Collection and SLAM Implementation:** Collected point cloud data from the LIDAR sensor and applied it to SLAM algorithms for accurate mapping and localization.
- **Implementation of GMapping SLAM Algorithm:** Implemented the GMapping algorithm using a Rao-Blackwellized particle filter, incorporating adaptive techniques and selective resampling to enhance localization accuracy and improve map quality.
- **KITTI Dataset Conversion and Integration:** Converted the KITTI dataset into a ROS2 bag file and integrated it with the GMapping algorithm, optimizing sensor data processing for improved SLAM performance.
- **Development of Object Tracking System:** Developed and integrated computer vision techniques for object tracking using a stereo camera. This system was successfully deployed in hardware to enhance the robot's obstacle avoidance capabilities.

His contributions were nothing short of outstanding. He consistently demonstrated a high level of professionalism, critical thinking, and adaptability in approaching complex challenges. I am confident that Abhisek will bring the same level of dedication, enthusiasm, and expertise to any organization or academic program he chooses to pursue.

Thank you for considering my recommendation. I am certain that Abhisek will make a valuable addition to any team or program he becomes a part of. Should you have any queries, you can reach me at ramprasad@iitbbs.ac.in / +91-674-713-5789.

Sincerely

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